CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 18: Constructions

Topic-wise short notes with solved examples

Ruler-and-compass constructions of angles and of quadrilaterals, parallelograms, rectangles, rhombuses and squares from given measurements.

1 Basic constructions

- Use only a ruler and compasses unless a protractor is allowed. Keep all arcs visible.
- **Perpendicular bisector of a segment AB**: with centres A and B and the same radius (more than half of AB), draw arcs on both sides. Join the two points where the arcs cross.
- **Bisector of an angle**: with the vertex as centre, cut both arms. From these two points draw equal arcs to cross. Join the vertex to the crossing point.
- **Angle of $60°$**: with centre A draw an arc cutting the line at P. With centre P and the same radius, cut the arc at Q. Then $\angle QAP = 60°$.

Angle	How to construct
$60°$	One step of the radius along the arc
$120°$	Two steps of the radius along the arc
$90°$	Bisect the angle between $60°$ and $120°$
$30°$	Bisect $60°$
$45°$	Bisect $90°$
$75°$	Bisect the angle between $60°$ and $90°$
$105°$	Bisect the angle between $90°$ and $120°$
$135°$	$90° + 45°$

2 Before constructing a quadrilateral

- Draw a **rough sketch** and mark the given measurements.
- A quadrilateral needs **five** independent measurements. Special quadrilaterals need fewer because their properties give the rest.

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Figure	Measurements needed
Quadrilateral	5
Parallelogram	3
Rectangle	2
Rhombus	2
Square	1

- Usually one triangle of the figure is constructed first; the fourth vertex is then found with arcs.

3 Quadrilaterals

Given	Method
4 sides and 1 diagonal	Draw the diagonal; make a triangle on each side of it with the given sides
4 sides and 1 angle	Draw two sides with the angle between them; cut arcs for the other two sides
3 sides and 2 included angles	Draw the middle side; make the angles at its ends; cut off the other two sides
3 sides and 2 diagonals	Draw the triangle with one side and the two segments from its ends; complete with arcs

Example 3.1 Construct quadrilateral ABCD with $AB = 5$ cm, $BC = 4$ cm, $CD = 4.5$ cm, $DA = 3.5$ cm and $BD = 6$ cm.

SOLUTION

Draw $BD = 6$ cm. With centres B and D and radii 5 cm and 3.5 cm, draw arcs meeting at A.

On the other side of BD, with centres B and D and radii 4 cm and 4.5 cm, draw arcs meeting at C.

Join AB, AD, BC and CD

Example 3.2 Construct quadrilateral PQRS with $PQ = 5$ cm, $QR = 4$ cm, $RS = 3.5$ cm, $\angle Q = 60^\circ$ and $\angle R = 90^\circ$.

SOLUTION

Draw $QR = 4$ cm. At Q construct 60° and cut off $QP = 5$ cm.

At R construct 90° and cut off $RS = 3.5$ cm.

Join PS

4 Parallelograms

- Use: opposite sides are equal and parallel; the diagonals bisect each other.

Given	Method
2 adjacent sides and the included angle	Draw the two sides with the angle; complete with arcs equal to the opposite sides
2 adjacent sides and a diagonal	Draw the triangle formed by them; complete with arcs
2 diagonals and the angle between them	Draw one diagonal and mark its mid-point O; draw the other through O at the given angle, half on each side
1 side and 2 diagonals	Draw the triangle with the side and the two half-diagonals; extend each half-diagonal by its own length

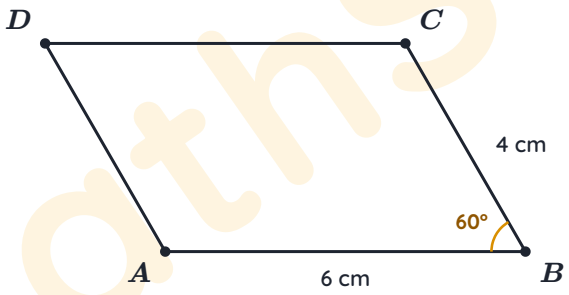
Example 4.1 Construct parallelogram ABCD with $AB = 6$ cm, $BC = 4$ cm and $\angle B = 60^\circ$.

SOLUTION

Draw $AB = 6$ cm. At B construct 60° and cut off $BC = 4$ cm.

With centre A and radius 4 cm, and centre C and radius 6 cm, draw arcs meeting at D.

Join AD and CD



Draw AB and BC first, then find D with two arcs

Example 4.2 Construct a parallelogram with diagonals 8 cm and 6 cm that meet at 60°.

SOLUTION

Draw $AC = 8$ cm and mark its mid-point O. Through O draw a line making 60° with AC.

On this line mark $OB = OD = 3$ cm on opposite sides of O.

Join AB, BC, CD and DA

5 Rectangles

Given	Method
Length and breadth	Draw the length; construct 90° at both ends; cut off the breadth on each
One side and a diagonal	Draw the side AB; construct 90° at B; from A cut this line with the diagonal as radius

Example 5.1 Construct rectangle ABCD with $AB = 6$ cm and $AC = 7.5$ cm.

SOLUTION

Draw $AB = 6$ cm. At B construct 90°. With centre A and radius 7.5 cm, cut the perpendicular at C. With centre A and radius BC, and centre C and radius 6 cm, draw arcs meeting at D.

Join AD and CD. (BC will measure 4.5 cm.)

6 Rhombuses

Given	Method
Side and one angle	Draw two sides with the angle; complete with arcs equal to the side
Side and one diagonal	Draw the diagonal; from both ends draw arcs equal to the side, on both sides of it
Two diagonals	Draw one diagonal; construct its perpendicular bisector; mark half of the other diagonal on each side

Example 6.1 Construct a rhombus with diagonals 8 cm and 6 cm.

SOLUTION

Draw $AC = 8$ cm and its perpendicular bisector, meeting AC at O.

Mark $OB = OD = 3$ cm on the bisector.

Join AB, BC, CD and DA. Each side will measure 5 cm.

7 Squares

Given	Method
Side	Draw the side; construct 90° at both ends; cut off the side on each; join
Diagonal	Draw the diagonal; construct its perpendicular bisector; mark half the diagonal on each side of the mid-point

Example 7.1 Construct a square with diagonal 6 cm.

SOLUTION

Draw $AC = 6$ cm and its perpendicular bisector through O.

Mark $OB = OD = 3$ cm on it and join AB, BC, CD and DA

8 Common mistakes

- Starting without a rough sketch.
- Using a protractor where a compass construction is required.
- Erasing the construction arcs.
- Marking the full diagonal on each side of O instead of half.
- Drawing the second triangle on the same side of the diagonal as the first.
- Not checking that the first triangle is possible (sum of two sides greater than the third).

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